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Incubator tray and a method of incubating

Abstract

An incubator tray includes a body that defines: (i) a first surface and a second surface that defines a sealed void therebetween; (ii) an inlet that permits inflow of fluid into the void defined between the first and second surfaces of the body; (iii) an outlet that permits outflow of fluid from the void defined between the first and second surfaces of the body; (iv) a plurality of bores that extend between the first and second surfaces, through the void defined between the first and second surfaces; and (v) a temperature sensor associated with each bore.

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Background/Summary

BACKGROUND

(1) The present invention relates to an incubator tray. More specifically, the present invention relates to a tray with bores/receptacles for receiving eggs and sensors for regulating incubation of the eggs. Furthermore, the present invention relates to a method of incubating eggs using an incubator tray with multiple sensors.

(2) Incubator trays that: define receptacles for receiving material/objects; are hollow; and facilitate circulation of temperature-controlled fluid within the trays to maintain material/objects supported on the trays at a constant temperature, are known. Such trays are described in EP1,266,691 “Temperature-controlled device and method suitable for spectroscopic analysis”, WO01/25866 “A system for controlling laboratory sample temperature and a thermal tray for use in such system”, WO2005/097325 “Autonomous device with active temperature regulation”, U.S. Pat. No. 3,556,731 “Automated incubation apparatus”, U.S. Pat. No. 4,195,131 “Environmentally controlled unit”, U.S. Pat. No. 4,256,697 “Blood incubator device”, U.S. Pat. No. 4,501,719 “Tray apparatus for freeze-drying biologicals having a predetermined unit dosage”, U.S. Pat. No. 4,735,778 “Microtiter plate”, U.S. Pat. No. 4,824,791 “Thermostated cuvette set”, U.S. Pat. No. 4,950,608 “Temperature regulating container”, U.S. Pat. No. 5,435,378 “Apparatus for accurately heating and cooling articles”, U.S. Pat. No. 5,475,610 “Thermal cycler for automatic performance of the polymerase chain reaction with close temperature control”, U.S. Pat. No. 5,484,015 “Cold plate and method of making same”, U.S. Pat. No. 5,504,007 “Rapid thermal cycle apparatus”, U.S. Pat. No. 6,558,947 “Thermal cycler”, US2006/0199261 “Holder and method for cooling or heating samples”, US2014/0273182 “Device for carrying out chemical or biological reactions” and US2015/0352553 “Methods and compositions for rapid thermal cycling”.

(3) However, prior art incubator trays do not: provide for ventilation around the material/objects received within the receptacles. More particularly, prior art incubators do not (in use) induce flow of air around material/objects received within the receptacles through venturi effect; or ensure close proximity of the sensors on the tray to the material/objects received within the receptacles.

(4) It is an object of this invention to address these drawbacks primarily by providing: sensors on protrusions within the receptacles; and modifying the receptacles to form bores that, in use, define open opposed axial ends with varying cross-sectional surface areas, thereby to induce a pressure differential along each bore (i.e. across the tray) when air flows along the tray.

SUMMARY OF THE INVENTION

(5) According to a preferred embodiment of a first aspect of the invention, there is provided an incubator tray that includes a body that defines: a first surface and a second surface that defines a sealed void therebetween; an inlet that, in use, permits inflow of fluid into the void defined between the first and second surfaces of the body; an outlet that, in use, permits outflow of fluid from the void defined between the first and second surfaces of the body; a plurality of bores that extend

between the first and second surfaces, through the void defined between the first and second surfaces, wherein: each bore is open at both axial ends; and a first axial end of the bore proximal the first surface is larger in cross-sectional area than the cross-sectional area of a second axial end of the bore proximal the second surface; and a temperature sensor associated with each bore, such that, in use, an object to be incubated that is: undersized relative to the cross-sectional area of the first axial end of a bore; and oversized relative to the cross-sectional area of the second axial end of the bore,

may be supported on the body, within a bore, with the open axial ends of the bore permitting ventilation through the bore, past the object.

(6) Typically, in respect of each bore, the body defines at least three protrusions that are spaced from each other and that protrude radially into the bore, such that, in use, an object supported on the body, within the bore, is supported on the protrusions associated with the bore.

(7) Generally, each protrusion extends at least partially along the bore, from the first surface to the second surface.

(8) Preferably, in respect of each bore, the temperature sensor is disposed on a protrusion.

(9) Typically, the incubator tray further includes: a heart rate sensor; a movement sensor; and a carbon dioxide sensor; and a sonic, magnetic, electrostatic or electromagnetic transmitter for transmitting waves to an object supported on the body, within a bore, associated with each bore.

(10) Generally, in respect of each bore, the heart rate sensor, movement sensor and carbon dioxide sensor are disposed within a protrusion.

(11) A heating element may be associated with each bore, which heating element is controlled by the temperature sensor associated with the associated bore.

(12) Preferably, the heating element comprises a flexible circuit including a plurality of resistors.

(13) Typically, the flexible circuit is in the form of a loop that extends about the radial perimeter of the bore.

(14) Generally, the incubator tray further includes a processor that is in communication with each flexible circuit within the tray.

(15) Preferably, an egg is supported within each bore and a blanket is laid over the eggs.

(16) Typically, the incubator tray further includes a diffuser associated with each of the inlet and the outlet to diffuse fluid that is entering and exiting the void defined between the first and second surfaces of the body.

(17) Generally, the incubator tray further includes a cover that covers: the first surface of the body, the bore and the flexible circuits.

(18) According to a second aspect of the invention, there is provided a method of incubating eggs on an incubator tray according to the first aspect of the invention, which method includes the steps of: orienting the incubator tray with the first surface of the body operatively above the second surface of the body; supporting an egg on the incubator tray, within a bore; passing air over the first and second surfaces of the body; regulating the temperature of the incubator tray by: monitoring the temperature of the egg using the temperature sensor; and adjusting the flow of fluid within the void defined between the first and second surfaces of the body via the inlet and outlet; monitoring: the temperature of the egg using the temperature sensor; the heart rate within the egg using the heart rate sensor; movement within the egg using the movement sensor; and the carbon dioxide level proximal the surface of the egg using the carbon dioxide sensor; and in respect of each egg, triggering an event should any two or more of the following conditions be met: the temperature sensor senses the egg temperature outside of a predetermined temperature range; the heart rate sensor senses a heart rate within the egg outside of a predetermined heart rate range; and the movement sensor senses movement within the egg outside of a predetermined range of activity level.

(19) Typically, the method of incubating eggs further includes the step of passing air over the first

surface of the body.

(20) Generally, the trigger event includes one or more of: communicating an alert that the embryo within the egg is dead; ceasing electricity supply to sensors associated with the egg; varying the temperature of the incubator tray; varying airflow over the first surface of the body; and varying the power output of the heating element associated with the bore in which the egg is supported.

Description

BRIEF DESCRIPTION OF THE DRAWINGS

(1) The invention will now be described in more detail, by way of example only, with reference to the accompanying drawings in which:

(2) FIG. 1 is a top view of an incubator tray according to a first embodiment of the invention;

(3) FIG. 2 is a cut-away perspective view of the incubator tray in FIG. 1 along line A-A;

(4) FIG. 3 is a cross-sectional side view of the incubator tray in FIG. 1 along line A-A;

(5) FIG. 4 is a cross-sectional side view of a bore defined by the incubator tray in FIG. 1;

(6) FIG. 5 is an exploded perspective view of an incubator tray according to an alternative embodiment of the invention;

(7) FIG. 6 is a cross-sectional side view of the incubator tray in FIG. 5 along line A-A; and

(8) FIG. 7 is a bottom view of the incubator tray in FIG. 5.

DESCRIPTION OF THE INVENTION

(9) With reference to FIGS. 1 to 4 of the drawings, a first embodiment of an incubator tray **10** comprises a body **12** that defines a first surface **14** and a second surface **16** and a plurality of bores **18** extending between the first and second surface **14** and **16**.

(10) The first and second surfaces **14** and **16** are generally planar. Importantly, the first and second surfaces **14** and **16** are: spaced from each other; and sealably connected to each other about their peripheral edge **22** and about the bores **18**,

to form a sealed void **20** between the first and second surfaces **14** and **16**.

(11) An inlet **24** permits inflow of temperature-regulated fluid into the void **20** defined between the first and second surfaces **14** and **16** of the body **12**, and an outlet **26** permits outflow of fluid from the void **20** defined between the first and second surfaces **14** and **16** of the body **12**.

(12) Each bore **18** extend between the first and second surfaces **14** and **16**, through the void **20** defined between the first and second surfaces **14** and **16**. Turning to FIG. 3, it will be appreciated that each bore **18** is open at both axial ends, albeit that a first axial end **18a** of the bore **18** proximal the first surface **14** is larger in cross-sectional area than the cross-sectional area of a second axial end **18b** of the bore **18** proximal the second surface **16**.

(13) The cross-sectional area of the first axial end **18a** of the bore **18** is oversized relative to an object (such as an egg) (not shown) to be supported on the body **12**, within the bore **18**, whereas the cross-sectional area of the second axial end **18b** of the bore **18** is undersized relative to such object.

(14) In respect of each bore **18**, the body **12** defines at least three protrusions **28** that are spaced from each other and that protrude radially into the bore **18**. In use, an object supported on the body **12**, within the bore **18**, is supported on the protrusions **28** associated with the bore **18** in a similar way that a tripod supports an object. FIGS. 1 to 4 show each protrusion **28** extending along its associated bore **18** from the first surface **14** of the body **12** to the second surface **16** of the body **12**. It will be appreciated that each protrusion **28** may extend along only a portion of the associated bore **18**.

(15) Furthermore, the set of three protrusions **28** associated with each bore **18**, in use, space an object supported within the bore **18** from the radial wall of the bore **18**, providing an annular gap (preferably, 3 mm in width) therebetween to facilitate ventilation through the bore **18**, past the object. This annular gap also permits radiation of heat from the body **12** to the object supported

within the bore **18**.

(16) A temperature sensor **30**, a heart rate sensor **32**, a movement sensor **34** and a carbon dioxide sensor **36** are associated with each bore **18**, disposed on a protrusion **28** extending along the bore **18**. Disposing the sensors **30**, **32**, **34** and **36** on the protrusions **28** ensures that the sensors **30**, **32**, **34** and **36** are in close contact with/proximity to objects supported on the body **12**, within the bores **18**. Alternatively, the heart rate sensor **32**, movement sensor **34** and carbon dioxide sensor **36** may be disposed within a protrusion **28**.

(17) Better to regulate the temperature objects supported on the body **12**, within the bores **18**, a heating element **38** is associated with each bore, which heating element **38** is regulated by the temperature sensor **30**.

(18) Optionally, the body **12** further defines a deflector (not shown) associated with each bore **18**, which deflector protrudes from one of the first or second surfaces **14** and **16** to divert air that, in use, flows along the first or second surface **14** and **16** through the bore **18**.

(19) Further optionally, a porous blanket (not shown) may, in use, be laid over the tray **12** (with objects supported on the body **12**, within the bores **18**).

(20) FIGS. **5** to **7** show an alternative embodiment of an incubator tray **110**, which comprises: a body **112** that defines a first surface **114** and a second surface **116** and a plurality of bores **118** extending between the first and second surface **114** and **116**; a sonic (whether audible or not), magnetic, electrostatic or electromagnetic transmitter **140** for transmitting waves to an object (e.g. an egg) supported on the body **112**, within a bore **118**.

(21) The transmitter **140** is in the form of a flexible circuit, preferably with a plurality of resistors, which flexible circuit is in the form of a loop that extends about the radial perimeter of the bore **118**. In use, the transmitter **140** may be used to heat the object supported on the body **112**.

(22) FIG. **7** also shows a diffuser **142** associated with each of the incubator tray **110** inlet **124** and outlet **126**, which diffuser **142** diffuses fluid that is entering and exiting the void **120** defined between the first and second surfaces **114** and **116** of the body **112**. Typically, the void **120** defined between the first and second surfaces **114** and **116** of the body **112** is designed with inlet and exit manifolds and other formations to distribute fluid flow evenly through the void **120**.

(23) Furthermore, a cover **144** overlies the first surface **114** of the body **112**, the bore **118** and the flexible circuits of the transmitter **140**.

(24) The incubator tray **10** and **110** may further include a processor (not shown) that is in communication with the temperature sensors **30**, heart rate sensors **32**, movement sensors **34**, carbon dioxide sensor **36** and/or transmitter **140**. FIG. **5** shows PCB strips **146** that electronically connect the temperature sensors **30**, heart rate sensors **32**, movement sensors **34**, carbon dioxide sensor **36** and/or transmitter **140**. The processor may, in turn be in communication (whether wired or wireless) with a primary controller.

(25) The invention further extends to a method of incubating eggs on an incubator tray **10**, which method includes the steps of: Orienting the incubator tray **10** with the first surface **14** of the body **12** operatively above the second surface **16** of the body **12**. Supporting an egg (not shown in FIGS. **1** to **4**, but shown in FIGS. **5** and **6**) on the incubator tray **10**, within each of the bores **18**. Passing air over the first and second surfaces **14** and **16** of the body **12**. It will be appreciated that, since the unobstructed surface area of the opening at the first axial end **18a** of the bore **18** is less than the surface area of the second axial end **18b** of the bore **18**, airflow over the first and second surfaces **14** and **16** of the body **12** induces a pressure differential along the bore **18**, which pressure differential induces flow of air along the bore from the first axial end **18a** of the bore **18** to the second axial end **18b** of the bore. The flow of air that causes this pressure differential along the bores **18** is typically called a venturi effect. Such air flow along the bore **18** removes carbon dioxide, humidity and heat from around the egg. Regulating the temperature of the incubator tray **10** by: monitoring the temperature of the egg using the temperature sensor **26**; and adjusting the flow of fluid within the void **20** defined between the first and second surfaces **14** and **16** of the

body **12** via the inlet and outlet **24** and **26**; Passing air over the first surface **14** of the body **12** and over the eggs and permitting air to pass along the annular gaps between the radial walls of the bores **18** and the eggs supported therein. In respect of each bore **18** and its associated egg, monitoring: temperature of the egg using the temperature sensor **30**; heart rate within the egg using the heart rate sensor **32**; movement within the egg using the movement sensor **34**; and the carbon dioxide level proximal the surface of the egg using the carbon dioxide sensor **36**. In respect of each bore **18** and its associated egg, triggering an event should any two or more of the following conditions be met: the temperature sensor **30** senses the egg temperature outside of a predetermined temperature range; the heart rate sensor **32** senses a heart rate within the egg outside of a predetermined heart rate range; and the movement sensor **34** senses movement within the egg outside of a predetermined range of activity level.

(26) The event triggered could include any one or more of: communicating an alert that the embryo within the egg is dead, which signal may be communicated to a remote device; ceasing electricity supply to sensors **30**, **32**, **34** and **36** associated with the egg; varying the temperature of the incubator tray **10** by adjusting fluid flow within the void **20** defined between the first and second surfaces **14** and **16** of the body **12** via the inlet and outlet **24** and **26**; varying airflow over the first surface **14** of the body **12**; and varying the output of the heating element **38** associated with the bore **18** within which the egg is supported.

(27) Feedback from the carbon dioxide sensor **36** exclusively may also be used to regulate airflow over the incubator tray **10** and over the eggs.

(28) It will be appreciated that air ventilation around the egg is facilitated by: the through-bores **18** that are open at both axial ends; and the protrusions **28** associated with each bore **18**. Furthermore, it will be appreciated that the protrusions **28** ensure close contact/proximity between the sensors on the incubator tray **12** and the eggs supported on the body **12**, within the bores **18**.

Claims

1. An incubator tray includes a body that defines: a first surface and a second surface that defines a sealed void therebetween; an inlet that permits fluid communication into the void defined between the first and second surfaces of the body; an outlet that permits fluid communication from the void defined between the first and second surfaces of the body; a plurality of bores that extend between the first and second surfaces, through the void defined between the first and second surfaces, wherein: each bore is open at both axial ends; and a first axial end of the bore proximal the first surface is larger in cross-sectional area than the cross-sectional area of a second axial end of the bore proximal the second surface; and a temperature sensor associated with each bore, such that an object to be incubated that is: undersized relative to the cross-sectional area of the first axial end of a bore; and oversized relative to the cross-sectional area of the second axial end of the bore, is supportable on the body, within the bore, with the open axial ends of the bore permitting ventilation through the bore, past the object.

2. The incubator tray according to claim 1, wherein, in respect of each bore, the body defines at least three protrusions that are spaced from each other and that protrude radially into the bore, such that an object supported on the body, within the bore, is supported on the protrusions associated with the bore.

3. The incubator tray according to claim 2, wherein, each protrusion extends along at least a portion of a length of the bore.

4. The incubator tray according to claim 3, wherein, in respect of each bore, the temperature sensor is disposed on a protrusion.

5. The incubator tray according to claim 4, further including: a heart rate sensor; a movement sensor; a carbon dioxide sensor; and a sonic, magnetic, electrostatic or electromagnetic transmitter for transmitting waves to an object supported on the body, within a bore, associated with each bore.

6. The incubator tray according to claim 5, wherein, in respect of each bore, the heart rate sensor, movement sensor and carbon dioxide sensor are disposed within a protrusion.
 7. The incubator tray according to claim 6, wherein a heating element is associated with each bore, which heating element is controlled by the temperature sensor associated with the associated bore.
 8. The incubator tray according to claim 7, wherein the heating element comprises a flexible circuit including a plurality of resistors.
 9. The incubator tray according to claim 8, wherein the flexible circuit is in the form of a loop that extends about the radial perimeter of the bore.
 10. The incubator tray according to claim 9, further including a processor that is in communication with each flexible circuit within the tray.
 11. The incubator tray according to claim 10, wherein an egg is supported within each bore and a blanket is laid over the eggs.
 12. The incubator tray according to claim 11, further including a diffuser associated with each of the inlet and the outlet to diffuse fluid that is entering and exiting the void defined between the first and second surfaces of the body.
 13. The incubator tray according to claim 12, further including a cover that covers: the first surface of the body, the bore and the flexible circuits.
 14. A method of incubating eggs on an incubator tray described in claim 7, which method includes the steps of: orienting the incubator tray with the first surface of the body operatively above the second surface of the body; supporting an egg on the incubator tray, within a bore; passing air over the first and second surfaces of the body; regulating the temperature of the incubator tray by: monitoring the temperature of the egg using the temperature sensor; and adjusting the flow of fluid within the void defined between the first and second surfaces of the body via the inlet and outlet; monitoring: the temperature of the egg using the temperature sensor; the heart rate within the egg using the heart rate sensor; movement within the egg using the movement sensor; and the carbon dioxide level proximal the surface of the egg using the carbon dioxide sensor; and in respect of each egg, triggering an event should any two or more of the following conditions be met: the temperature sensor senses the egg temperature outside of a predetermined temperature range; the heart rate sensor senses a heart rate within the egg outside of a predetermined heart rate range; and the movement sensor senses movement within the egg outside of a predetermined range of activity level.
 15. The method of incubating eggs according to claim 14 further including the step of passing air over the first surface of the body.
 16. The method of incubating eggs according to claim 15, wherein the trigger event includes one or more of: communicating an alert that the embryo within the egg is dead; ceasing electricity supply to sensors associated with the egg; varying the temperature of the incubator tray; varying airflow over the first surface of the body; and varying the power output of the heating element associated with the bore in which the egg is supported.
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